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1 # AuditAd
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3 AuditAd is a hybrid mobile/web app built with React, TypeScript, Vite, and Capacitor.
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5 # AuditAd Privacy Dashboard
6
7 AuditAd is a high-performance local DNS threat optimizer and privacy audit utility for
  Android. It operates as a local loopback to monitor, audit, and block intrusive
  telemetry, ad-frames, and cross-app profiling hooks without routing your data to external
  servers.
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9 ## Features & Tabs
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11 ### 1. Diagnostic Center
12 The heartbeat of AuditAd. It displays a live stream of DNS queries intercepted by the
  local VPN loopback (Port 53).
13 - **Live Logging**: Real-time visibility into which apps are making network requests and
  where they are going.
14 - **Categorization**: Automatically flags queries as Ads, Analytics, Trackers, or
  Allowed.
15 - **Latency Tracking**: Measures the overhead of on-device filtration (typically <
  1.8ms).
16 - **Control**: Pause the stream or clear logs at any time to focus on specific app
  activity.
17
18 ### 2. Bloom Filter Playground
19 A deep-dive visualization into AuditAd's zero-knowledge matching engine.
20 - **Simulation**: Understand how the app checks hostnames against 150,000+ domains in
  sub-millisecond time.
21 - **Interactive Math**: Adjust parameters like Bit Array Size (m), Tracked Domains (n),
  and Hash Functions (k) to see the effect on the False Positive Probability.
22 - **Memory Map**: A visual 64-bit memory grid showing how trackers "flip bits" in kernel
  memory.
23
24 ### 3. Package Auditor Engine
25 A security scanner for Android Manifest files and installed packages.
26 - **Manifest Deep Audit**: Paste an `AndroidManifest.xml` and use Gemini AI to generate
  an instant privacy risk scorecard.
27 - **Risk Indexing**: Calculates a threat score (0-100) based on excessive permissions and
  background intent receivers.
28 - **Package Catalogue**: Browse "installed" apps (like *Shadow Legends: RPG* or
  *ChatterBox Ultimate*) to see their specific tracking liabilities and Play Store links.
29
30 ### 4. Account Janitor
31 A direct deep-linking helper to guide users through complex corporate privacy settings.
32 - **Interactive Tutorials**: Step-by-step overlays for Google and Meta (Facebook) privacy
  controls.
33 - **Direct Routing**: One-click handles to launch native Android settings or web-based
  privacy centers.
34 - **Advertising ID Reset**: Guides you to wipe your hardware-bound Advertising ID to
  neutralize cross-app profiling.
35
36 ### 5. Private DNS Settings
37 Educational center for configuring Android's native Private DNS (TLS/HTTPS).
38 - **Battery Optimization**: Explains why Native DNS is the most efficient defense (0%
  battery drain compared to local VPN loops).
39 - **Presets**: Copiable hostnames for AdGuard, Cloudflare, and Quad9.
40 - **Setup Guide**: Visual walkthrough for Android 9+ system settings.
41
42 ## Native Enhancements
43 - **Ghost App Prevention**: A specialized `AppCheckPlugin` uses explicit `PackageManager`
  flag filters (`MATCH_DEFAULT_ONLY`) to prevent stale metadata from showing uninstalled
  apps as "installed."
44 - **Capacitor 6 Bridge**: Seamless integration between the React-based UI and native
  Android system APIs.
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46 ## Build & Deployment
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48 ### Prerequisites
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```
49 - Node.js & npm
50 - Android Studio & Android SDK
51
52 ### Running Locally
53 1. Install dependencies:
54   ```bash
55   npm install
56   ```
57 2. Set the `GEMINI_API_KEY` in your environment (for the Auditor Engine).
58 3. Build web assets and sync to Android:
59   ```bash
60   npm run build
61   npx cap sync android
62   ```
63 4. Run via Android Studio or the CLI:
64   ```bash
65   npx cap open android
66   ```
67
68 ## License
69 SPDX-License-Identifier: Apache-2.0
70
71
72 ## Project Overview
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74 - **UI:** Built using React 19 + TypeScript with Vite and Tailwind CSS.
75 - **Mobile shell:** Capacitor Android integration provides the Android wrapper and access
    to native device features.
76 - **Android native code:** Custom Capacitor plugin code is implemented in Java under
    `android/app/src/main/java/...`.
77 - **Data persistence:** The app currently uses browser `localStorage` for simple state
    persistence and settings such as theme preference and workspace state.
78 - **No Kotlin UI layer:** There are no Kotlin source files in this repository; the web UI
    is rendered from the React app, while Android native integration is Java-based.
79
80 ## Tech Stack
81
82 - `react` + `react-dom`
83 - `typescript`
84 - `vite`
85 - `tailwindcss`
86 - `@capacitor/core`
87 - `@capacitor/android`
88 - `express` and `dotenv` for local development server support
89
90 ## Run Locally
91
92 **Prerequisites:** Node.js
93
94 1. Install dependencies:
95   `npm install`
96 2. Set the `GEMINI_API_KEY` in `.env.local` to your Gemini API key.
97 3. Run the app:
98   `npm run dev`
99
100 ## Android
101
102 1. Build the web app and sync Capacitor:
103   `npm run android`
104 2. This runs the Capacitor sync and opens the Android project in Android Studio.
105
106 ## Notes
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108 - The current project does not include a dedicated mobile database layer like SQLite or
    Room.
109 - Persistence is handled with `localStorage` for session-specific values and simple
    settings.
110 - If you want to add richer device-side storage, you can integrate Capacitor Storage,
    SQLite, or an Android database solution later.
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